INCH-POUND

MIL-L-52430C
2 December 1991
SUPERSEDING
MIL-L-52430B
18 September 1975

MILITARY SPECIFICATION

LIGHTS, SIGNAL, SURVEYING:

2-INCH AND 5-INCH REFLECTORS,

BATTERY-POWERED, WITH CASES

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers two sizes of battery-powered surveying signal lights with cases.

1.2 Classification. Signal lights shall be the following sizes, as specified (see 6.2):

Size 1 - 2-inch reflector.
Size 2 - 5-inch reflector.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. The following specifications and standards form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2).

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: USA Belvoir Research, Development, and Engineering Center, ATTN: STRBE-TSE, Fort Belvoir, VA 22060-5606 by using the Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A

FSC 6675

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SPECIFICATIONS

FEDERAL

PPP-B-601	- Boxes, Wood, Cleated-Plywood.
PPP-B-636	- Boxes, Shipping, Fiberboard.

MILITARY

MIL-T-704	- Treatment and Painting of Materiel.
MIL-C-46168	- Coating, Aliphatic Polyurethane, Chemical Agent Resisting.

STANDARDS

MILITARY

MIL-STD-105	- Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129	- Marking for Shipment and Storage.
MIL-STD-130	- Identification Marking of US Military Property.
MIL-STD-889	- Dissimilar Metals.
MIL-STD-1190	- Minimum Guidelines for Level C Preservation, Packing and Marking.
MS15569	- Lamp, Incandescent G-6 Bulb, Double Contact, Bayonet Candelabra Base.

(Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from the Standardization Documents Order Desk, Building 4D, 700 Robbins Avenue, Philadelphia, PA 19111-5094.)

2.1.2 Government drawings. The following Government drawings form a part of this document to the extent specified herein. Unless otherwise specified, the issues shall be those in effect on the date of the solicitation.

DRAWINGS

ME

TA13215E7182	- Light, Signal, Surveying, 5-Inch Reflector.
TA13217E3730	- Light, Signal, Surveying, 2-Inch Reflector.
13217E3755	- Receptacle.
13217E3753	- Lamp.

(Copies of drawings, publications, and other Government documents required by contractors in connection with specific acquisition functions should be obtained from the USA Belvoir Research, Development, and Engineering Center, ATTN: STRBE-JDT, Fort Belvoir, VA 22060-5606.)

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2.2 Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are DOD adopted are those listed in the issue of the DODISS cited in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS are the issues of the documents cited in the solicitation (see 6.2).

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

D 3953 - Strapping, Flat Steel and Seals.
D 4675 - Selection and use of Flat Strapping Materials.

(Application for copies should be addressed to the American society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

(Non-Government standards and other publications are normally available from the organizations that prepare or distribute the documents. These documents also may be available in or through libraries or other informational services.)

2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, (except for related associated detail specifications, specification sheets or MS standards), the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Description. The size 1 signal light and case shall be similar to TA13217E3730 and as specified herein. The receptacle for the size 1 signal light shall be similar to drawing 13217E3755. The size 2 signal light and case shall be similar to TA13215E7182 and as specified herein.

3.1.1 Drawings. The drawings forming a part of this specification are information drawings (see 6.4).

3.2 First article. Unless otherwise specified (see 6.2), a sample shall be subjected to first article inspection (see 6.3) in accordance with 4.3.

3.3 Materials. Materials shall be as specified herein and as shown on the applicable drawings. Materials not specified shall be selected by the contractor and shall be subject to all provisions of this specification.

3.3.1 Material deterioration prevention and control. The signal light shall be fabricated from compatible materials, inherently corrosion resistant or treated to provide protection against the various forms of corrosion and deterioration that may be encountered in any of the applicable operating and storage environments to which the signal light may be exposed.

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3.3.2 Dissimilar metals. Dissimilar metals shall not be used in intimate contact with each other unless protected against galvanic corrosion. Dissimilar metals and methods of protection are defined and detailed in MIL-STD-889.

3.3.3 Identification of materials and finishes. The contractor shall identify the specific material, material finish or treatment for use with component and subcomponent, and shall make information available upon request to the contracting officer or designated representative.

3.3.4 Recovered materials. For the purpose of this requirement, recovered materials are those materials which have been collected from solid waste and reprocessed to become a source of raw materials, as distinguished from virgin raw materials. The components, pieces and parts incorporated in the signal light may be newly fabricated from recovered materials to the maximum extent practicable, provided the signal light produced meets all other requirements of this specification. Used, rebuilt or remanufactured components, pieces and parts shall not be incorporated in the signal light.

3.4 Reflecting surface. The reflecting surfaces shall be plated, polished and coated to prevent oxidation. The reflecting surfaces shall have a coefficient of reflection of not less than 75 percent.

3.5 Accessories. Accessories shall consist of screws for fastening the light to a tripod and to another light and a carrying strap.

3.6 Case loading plan. A case loading plan shall be securely affixed inside the lid of the case and shall show the proper placing of light, accessories, replacement lamps, and arrangement of the carrying strap to permit carrying two lights with one strap, similar to information figures 1, 2 and 3 (see 6.5).

3.7 Light projection.

3.7.1 Size 1. When tested as specified in 4.5.2.2, the signal light shall project a beam of light that is focusable into a homogenous spot with no dark areas within a 9-inch diameter circle 5 feet from the lens.

3.7.2 Size 2. When tested as specified in 4.5.2.2, the signal light shall project a beam of light that is focusable into a homogenous spot with no dark areas within a 6-inch diameter circle 10 feet from the lens.

3.8 Lamps.

3.8.1 Size 1. Ten PR-2 lamps and ten PR-9 lamps shall be furnished in a package and included in the case. One PR-9 lamp shall be inserted in the signal light. The lamps shall be in accordance with drawing 13217E3753.

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3.8.2 Size 2. Two Trade No. 90 lamps, one Trade No. 304 lamp and one Trade No. 82 lamp shall be furnished and inserted in the lid of the case. One Trade No. 82 lamp shall be inserted in the signal light. The lamps shall be in accordance with MS15569.

3.9 Treatment and painting. Unless otherwise specified (see 6.2) the portions of the target light normally painted shall be painted with chemical agent resistant coating (CARC) conforming to MIL-C-46168 and applied in accordance with MIL-T-704, type F or G.

3.10 Identification marking. The signal lights shall be identified in accordance with MIL-STD-130 and the drawings referenced herein.

3.11 Workmanship. All parts, components and assemblies of the signal lights and cases shall be clean and free from harmful extraneous material. All edges shall be rounded or beveled.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements (examinations and tests) as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to ensure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items shall meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of ensuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling inspection, as part of manufacturing operations, is an acceptable practice to ascertain conformance to requirements, however, this does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance of defective material.

4.1.2 Component and material inspection. The contractor is responsible for ensuring that components and materials are manufactured, examined, and tested in accordance with referenced specifications and standards, as applicable.

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4.2 Classification of inspection. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.3).
b. Quality conformance inspection (see 4.4).
c. Inspection of packaging (see 4.6).

4.3 First article inspection.

4.3.1 Examination. The first article signal light shall be examined as specified in 4.5.1. Presence of one or more defects shall be cause for rejection.

4.3.2 First article test. The first article shall be tested as specified in 4.5.2. Failure of any test shall be cause for rejection.

4.4 Quality conformance inspection.

4.4.1 Sampling. Sampling for examination and test shall be in accordance with MIL-STD-105. Sample size shall be determined by using MIL-STD-105, table I and table II-A.

4.4.2 Examination. Samples selected as specified in 4.4.1 shall be examined for the defects specified in 4.5.1. Presence of one or more defects shall be cause for rejection of the lot.

4.4.3 Tests. Samples selected as specified in 4.4.1 shall be tested as specified in 4.5.2. Failure of any test shall be cause for rejection of the lot.

4.5 Inspection procedure.

4.5.1 Examination. The signal lights and cases shall be examined as specified herein for the following defects:

Major

101. Signal lights or cases not similar to drawings (see 3.1).
102. Material not as specified (see 3.3).
103. Materials are not corrosion resistant or treated to be made corrosion resistant for the applicable storage and operating environment (see 3.3.1).
104. Dissimilar metals as defined in MIL-STD-889 are not effectively insulated from each other (see 3.3.2).
105. Contractor does not have documentation available for identification of material, material finishes or treatments (see 3.3.3).
106. Used, rebuilt or remanufactured components, pieces, or parts incorporated in the signal light (see 3.3.4).

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107. Accessories not as specified (see 3.5).
108. Case loading plan not as indicated (see 3.6).
109. Lamps not as specified (see 3.8).
110. Treatment and painting not as specified (see 3.9).
111. Identification marking missing or illegible (see 3.10).
112. Workmanship not as specified (see 3.11).

4.5.2 Tests.

4.5.2.1 Reflective efficiency. The reflective efficiency of the reflectors shall be determined by the absorption of light of a reflected collimated beam or other suitable source. The loss of light in the process of reflection shall be determined by comparing the light reflected (in foot candles) with the light projected (in foot candles) upon the reflector surface and expressing the result as a percentage. A reflection coefficient less than 75 percent shall constitute failure of this test (see 3.4).

4.5.2.2 Light projection. A beam shall be projected from the signal light, powered by the nominal voltage of the bulb, to a screen that is perpendicular to the projected optical axis of the signal light. Nonconformance to 3.7.1 or 3.7.2, as applicable, shall constitute failure of this test.

4.6 Inspection of packaging.

4.6.1 Quality conformance inspection of pack.

4.6.1.1 Unity of product. For the purpose of inspection, a completed pack prepared for shipment shall be considered a unit of product.

4.6.1.2 Sampling. Sampling for examination shall be in accordance with MIL-STD-105.

4.6.1.3 Examination. Samples selected in accordance with 4.6.1.2 shall be examined for the following defects. Sample size shall be determined by using MIL-STD-105, table I and table II-A. A lot shall be accepted when zero defects are found and rejected when one or more defects are found.

113. Preservation not as specified for level A, B, or C (see 5.1).
114. Box closure not as specified for level A (see 5.1.1).
115. Signal lights of unlike description packed together for level A, B or C (see 5.2).
116. Packing not as specified for level A, B, or C (see 5.2).
117. Gross weight of the boxes exceeds 200 pounds for level A (see 5.2.1).
118. Strapping not as specified for level A (see 5.2.1).

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119. Quantities packed together exceed the size or weight limitation of the box for level B (see 5.2.2).
120. Marking illegible, incorrect or missing for level A, B, or C (see 5.3).

5. PACKAGING

5.1 Preservation. Preservation shall be level A or C, as specified (see 6.2).

5.1.1 Level A. Each case containing a signal light and accessories shall be placed in a close-fitting box conforming to PPP-B-636, W6C, style optional. Box closure shall be in accordance with method V of the appendix to the box specification.

5.1.2 Level C. Each case containing a signal light and accessories shall be preserved in a manner to afford protection against deterioration and damage during shipment from the contractor to the initial destination.

5.2 Packing. Packing shall be level A, B, or C, as specified (see 6.2).

5.2.1 Level A. Signal lights of like description, preserved as specified in 5.1, shall be packed together in close-fitting boxes conforming to PPP-B-601, overseas type, style optional. The gross weight of each box shall not exceed 200 pounds. Strapping shall conform to ASTM D 3953, type 1 or 2, zinc coated, size as applicable and ASTM D 4675.

5.2.2 Level B. Signal lights of like description, preserved as specified in 5.1, shall be packed together in close-fitting boxes conforming to PPP-B-636, V3c, V11c, V13c, or V15c, style optional, in quantities not to exceed the size or weight limitation of the box. Box closure and sealing shall be in accordance with the appendix to the box specification.

5.2.3 Level C. Signal lights of like description, preserved as specified in 5.1, shall be packed together in accordance with MIL-STD-1190.

5.3 Marking. In addition to any special marking specified in the contract or purchase order (see 6.2), marking for shipment and storage shall be in accordance with MIL-STD-129.

6. NOTES

(This section contains information of general or explanatory nature which is helpful, but is not mandatory.)

6.1 Intended use. The signal lights are intended for use as illuminated targets for triangulation surveying at night.

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6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of the specification.
b. Size of signal lights required (see 1.2).
c. Issue of DODISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see 2.1.1 and 2.2).
d. When a first article is required for inspection and approval and number of units required (see 3.2).
e. Level of preservation and level of packing required (see 5.1 and 5.2).
f. When painting shall be other than as specified (see 3.9).
g. Color required when other than is specified (see 3.9).
k. Any special marking (see 5.3).

6.3 First article. When a first article inspection is required, the item(s) should be a preproduction model. The first article should consist of one or more units. The contracting officer should include specific instructions in acquisition documents regarding arrangements for examinations, approval of the first article test results and disposition of the first articles. Invitation for bids should provide that the Government reserves the right to waive the requirement for samples for first article inspection to those bidders offering a product which has been previously acquired or tested by the Government, and that bidders offering such products, who wish to rely on such production or test, must furnish evidence with the bid that prior Government approval is presently appropriate for the pending contract. Bidders should not submit alternate bids unless specifically requested to do so in the solicitation.

6.4 Information drawings. TA13215E7182, TA13217E3730, drawings 13217E3755 and 13217E3753 are based on a design which has met performance requirements and is considered to be equivalent to that required by this specification. The drawings are furnished for information only. The Government assumes no responsibility for the accuracy of the drawings, and the use of this design does not relieve the contractor from complying with all requirements of this specification.

6.5 Information figures. Figures 1, 2, and 3 show loading plans and strap arrangements which have been found acceptable; however, the figures are included for illustration only and are not intended to preclude the furnishing of other plans which conform to the specification.

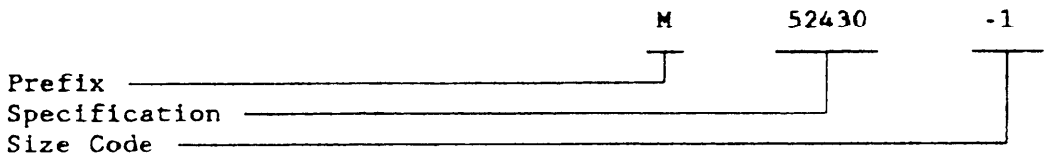
6.6 Subject (key word) listing.

Reflecting surface
Case loading
Light projection

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6.7 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

6.8 Part or identifying number (PIN). The part or identifying number (PIN) of signal lights covered by this specification shall be designated in the following form (see 6.2):



Example: M52430-1 is the PIN for a 2-inch reflector.

6.8.1 Size code. See table I.

TABLE I. Size code.

Code	Size	Description
1	1	2-inch reflector
2	2	5-inch reflector

Custodian:
Army - ME
Air Force - 99

Review activity:
Air Force - 82
DLA - GS

User activity:
Navy - MC

Preparing Activity:
Army - ME

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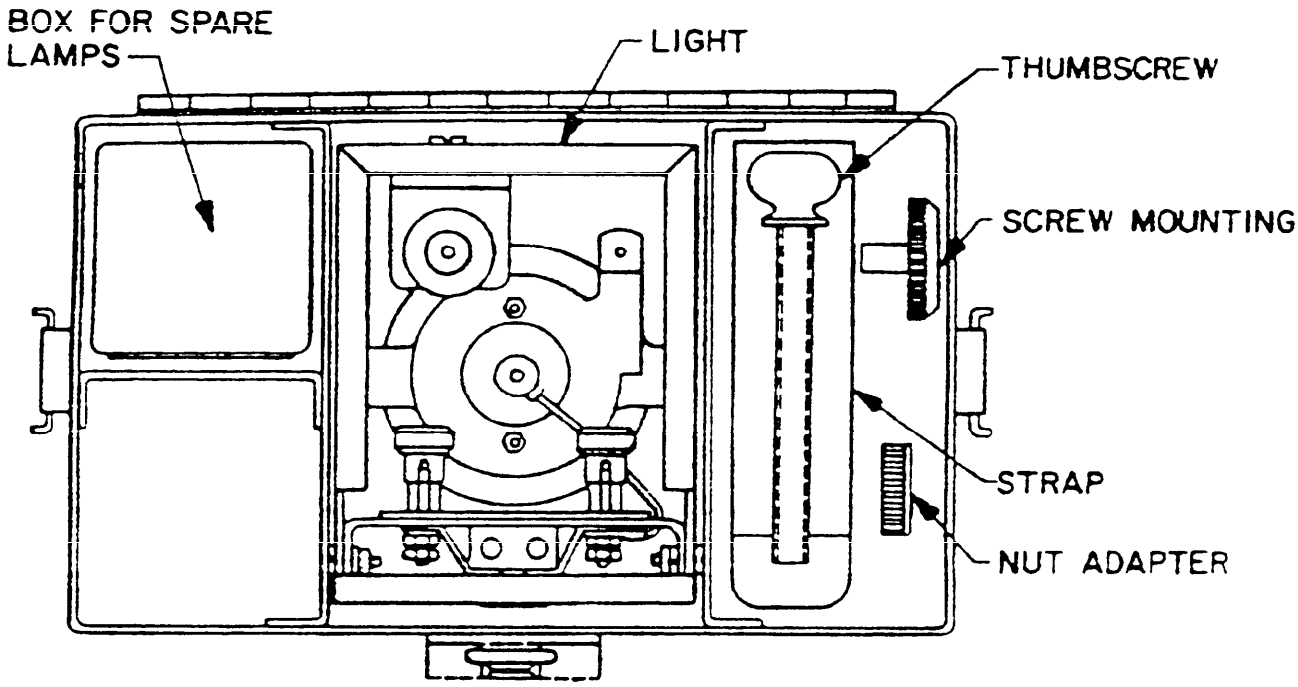
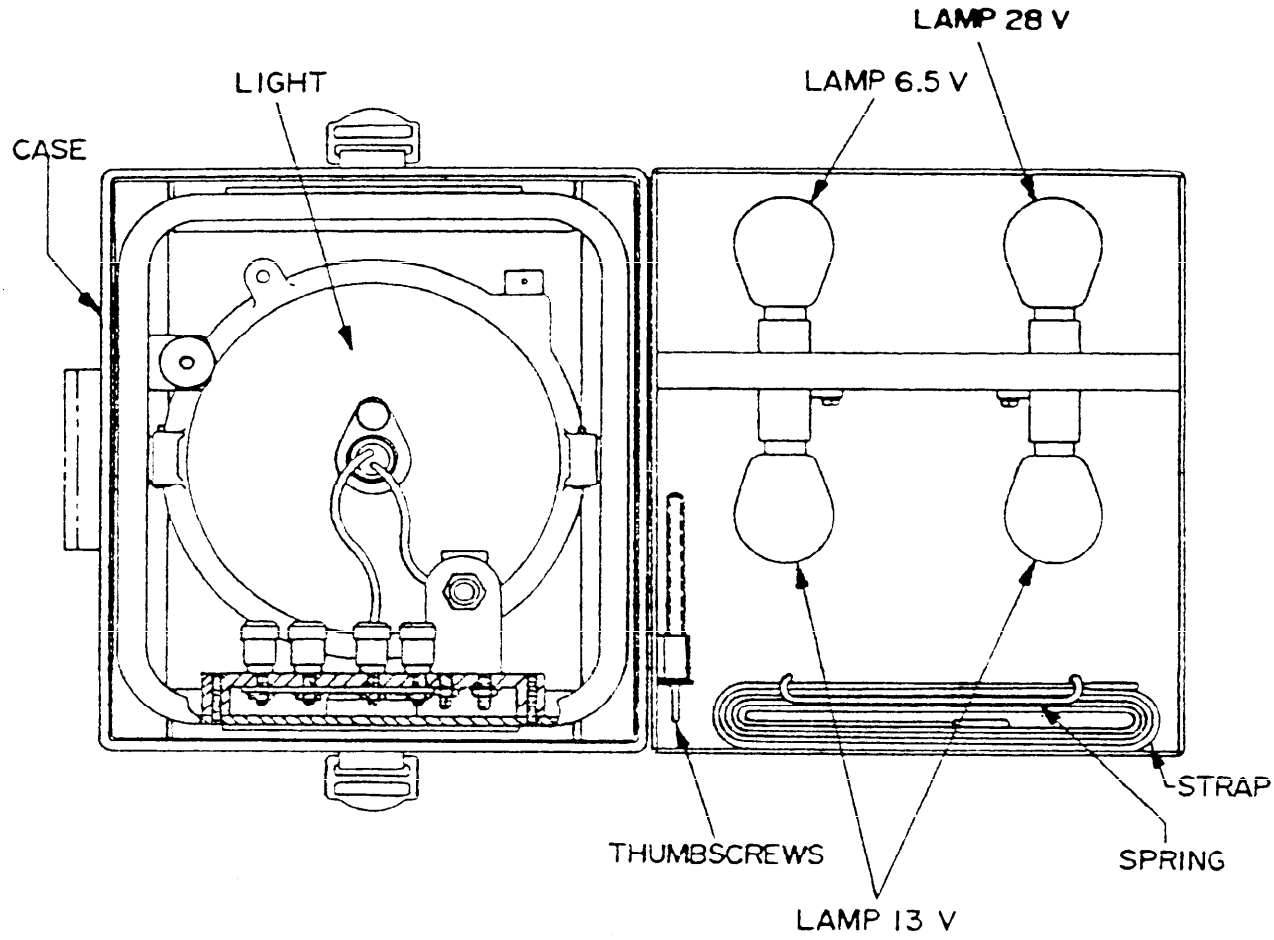


FIGURE 1. Surveying signal lamp, 2-inch reflector, loading plan.

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TOP VIEW WITH COVER IN 180° OPEN POSITION

FIGURE 2. Surveying signal light, 5-inch reflector, loading plan.

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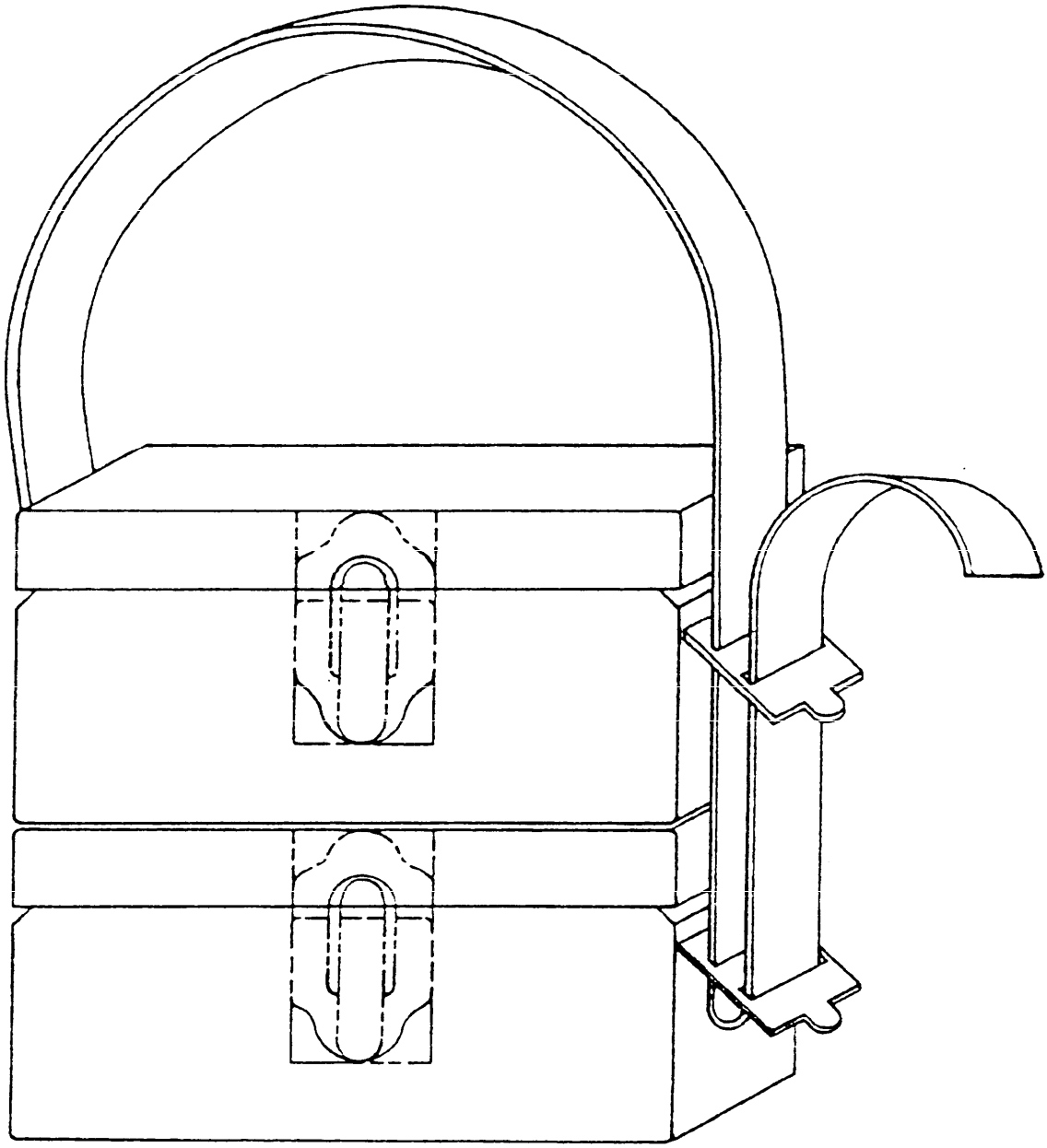


FIGURE 3. Surveying signal light, carrying strap arrangement for stacked lights.

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INSTRUCTIONS

1. The preparing activity must complete blocks 1, 2, 3, and 8. In block 1, both the document number and revision letter should be given.
2. The submitter of this form must complete blocks 4, 5, 6, and 7.
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NOTE: This form may not be used to request copies of documents, nor to request waivers, or clarification of requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements.

I RECOMMEND A CHANGE:	1. DOCUMENT NUMBER	2. DOCUMENT DATE (YYMMDD)
	MIL-L-52430C	91/12/02

3. DOCUMENT TITLE

Lights, Signal, Surveying: 2-Inch & 5-Inch Reflectors, Battery-Powered, with Cases

4. NATURE OF CHANGE *(Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)*

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME (Last, First, Middle Initial)	b. ORGANIZATION
c. ADDRESS (Include Zip Code)	d. TELEPHONE (Include Area Code) (1) Commercial (2) AUTOVON	e. DATE SUBMITTED (YYMMDD)

7. PREPARING ACTIVITY

a. NAME	b. TELEPHONE (Include Area Code) (1) Commercial	(2) AUTOVON
	(703) 664-5717	354-5717
c. ADDRESS (Include Zip Code) US Army Belvoir RDE Center ATTN: STRBE-TSE Ft. Belvoir, VA 22060-5606	IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT: Defense Quality and Standardization Office 5203 Leesburg Pike, Suite 1403, Falls Church, VA 22041-3466 Telephone (703) 756-2340 AUTOVON 289-2340	

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